

On the Use of Gröbner Bases in a Robotics Course

Manfred Husty[1(✉)] and Paul Zsombor-Murray[2]

[1] Unit Geometry and CAD, University of Innsbruck, Innsbruck, Austria
Manfred.Husty@uibk.ac.at
[2] CIM, McGill University, Montreal, Canada
paul@cim.mcgill.ca

Abstract. Most robotic mechanical systems consist of rigid links connected via revolute or prismatic joints. Therefore their kinematic behaviour can be described by nonlinear algebraic (polynomial) equations. These equations are called constraint equations and can be used to solve the most important kinematic problems like forward and inverse kinematics or singularities. Whereas methods from algebraic geometry to treat these equations are used in scientific papers, these often very elegant methods are rarely used in engineering courses. This paper reports on some of these methods and how they were used in a robotics course.

1 Introduction

The use of Gröbner bases in an engineering course needs some prerequisites and a certain sequence of contents. the first step is to introduce the rational representation of the Euclidean displacement, which is done using Study's kinematic mapping. This generally need no more than one lecture, because the students know already other representations like Euler angles, Rodrigues parameters or Euler parameters. The second step is a brief introduction into Gröbner bases and the notions of algebraic geometry which are linked to this tool. A very rough introduction to this mathematical concept done without any proof, needs no more than one lecture. Then the application of this tool can start. Within this paper due to the limitation of space only the basic ideas can be described and two out of many application will be shown. The paper is organized as follows: In Sect. 2 a brief introduction to the kinematic image space is given. Section 3 contains the basic definitions from algebraic geometry and Sect. 4 shows two examples where the previously developed concepts have been applied. These are: the general equation of the coupler curve of a four-bar mechanism and the direct kinematic of the Stewart platform. The paper finishes with a conclusion.

2 Rational Representation of Euclidean Displacements, Kinematic Mapping

This brief introduction to the concept of rational representation of Euclidean displacements is taken from our paper [4]. Euclidean rigid body displacements

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are often described by homogeneous 4×4 matrices $\mathbf{M}$, that act on a point $\mathbf{x}$ located in a moving frame according to $\mathbf{x}' = \mathbf{M}\mathbf{x}$. $\mathbf{x}'$ is the image point in a base frame, the lower right 3×3 sub matrix of $\mathbf{M}$ is a proper orthogonal matrix encoding the orientation of the moving fame with respect to the base frame and the first column of $\mathbf{M}$ contains the vector connecting the origins of both frames therefore representing the translational part of the transformation. Using Study's kinematic mapping κ (see [3,7]), the displacement given by $\mathbf{M}$ is mapped to a point $\mathbf{d} = [x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3]^T$ in a seven dimensional projective space P^7. These point coordinates are called the *Study parameters* of the displacement. They fulfill the quadratic condition

$$x_0y_0 + x_1y_1 + x_2y_2 + x_3y_3 = 0, \tag{1.1}$$

which is called *Study condition*. Its zero set is the *Study quadric* $S_6^2 \subset P^7$.

In the inverse kinematic mapping a point on S_6^2 minus the *exceptional three space* $\mathcal{E}$: $x_0 = x_1 = x_2 = x_3 = 0$ yields the matrix

$$\mathbf{M} := \kappa^{-1}(\mathbf{d}) = \frac{1}{\Delta}\begin{bmatrix} 1 & 0 & 0 & 0 \\ t_1 & x_0^2 + x_1^2 - x_3^2 - x_2^2 & -2x_0x_3 + 2x_2x_1 & 2x_3x_1 + 2x_0x_2 \\ t_2 & 2x_2x_1 + 2x_0x_3 & x_0^2 + x_2^2 - x_1^2 - x_3^2 & -2x_0x_1 + 2x_3x_2 \\ t_3 & -2x_0x_2 + 2x_3x_1 & 2x_3x_2 + 2x_0x_1 & x_0^2 + x_3^2 - x_2^2 - x_1^2 \end{bmatrix} \tag{1.2}$$

where $\Delta = x_0^2 + x_1^2 + x_2^2 + x_3^2$ and

$$\begin{aligned} t_1 &= 2x_0y_1 - 2y_0x_1 - 2y_2x_3 + 2y_3x_2, \\ t_2 &= 2x_0y_2 - 2y_0x_2 - 2y_3x_1 + 2y_1x_3, \\ t_3 &= 2x_0y_3 - 2y_0x_3 - 2y_1x_2 + 2y_2x_1. \end{aligned} \tag{1.3}$$

Properties of kinematic chains are described with respect to (arbitrarily chosen) base and end effector coordinate frames. Possible locations of the end-effector with respect to the base correspond to algebraic varieties described by sets of polynomial equations in $\mathbb{P}^7$. The representation of planar displacements can be obtained by setting $x_1 = x_2 = y_0 = y_3 = 0$ and taking the upper left 3×3 sub-matrix of Eq. 1.2.

3 Algebraic Geometry Basics

In this introduction only the most important definitions and theorems are given. It is based on the book *"Ideals, Varieties, and Algorithms"* by *David Cox, John Little* and *Donal O'Shea* [2], which is an excellent introduction to algebraic geometry. More detailed descriptions and also the proofs for the theorems can be found there.

Examples were computed using *Maple 18* with its package `Groebner`. This package contains the low-level commands which were enough to compute all examples of the course, the `PolynomialIdeals` package is newer and contains the more sophisticated commands. There are also other software packages like e.g. *Singular*, *Macaulay 2* or *Mathematica* for such computations.

In the following all algebraic equations are polynomials in the ring $\mathbf{K}[\underline{x}] = \mathbf{K}[x_1, \ldots, x_n]$ where $\mathbf{K}$ is a field like $\mathbb{Q}$ or $\mathbb{C}$.

Ideals and Affine Varieties

At first polynomial ideals are defined which are the basic objects for everything else.

Definition 1. A set $I \subseteq \mathbf{K}[\underline{x}]$ is called an **ideal** if the following conditions are fulfilled:

- $\forall\, f, g \in I : f + g \in I$
- $\forall\, f \in I$ and $\forall\, h \in \mathbf{K}[\underline{x}] : hf \in I$

It follows that almost all ideals are infinite sets of polynomials and cannot be written down as a whole. The sole exception is the ideal $\{0\}$ which is also a proper subset resp. subideal of all other ideals because 0 is contained in every ideal. There is also an ideal which is a proper superset resp. superideal of them, namely the ideal which contains the constant polynomial 1 and with it all polynomials of $\mathbf{K}[\underline{x}]$.

Using Definition 1 it is possible now to define the ideal generated by a set of given polynomials $f_1, \ldots, f_s$.

Definition 2. Let $f_1, \ldots, f_s$ be polynomials in $\mathbf{K}[\underline{x}]$. Then the set

$$\langle f_1, \ldots, f_s \rangle = \{g \in \mathbf{K}[\underline{x}] : g = \sum_{i=1}^{s} h_i f_i \;\; and \;\; h_1, \ldots, h_s \in \mathbf{K}[\underline{x}]\}$$

is the **ideal generated by** $f_1, \ldots, f_s$.

In kinematics of mechanical systems one is much more interested in the affine variety that is associated with an ideal. Its definition is given below

Definition 3. For a given ideal $I = \langle f_1, \ldots, f_s \rangle \subseteq \mathbf{K}[\underline{x}]$ the set

$$\mathbf{V}(I) = \{(a_1, \ldots, a_n) \in \mathbf{K}^n : f_i(a_1, \ldots, a_n) = 0 \;\; for\ all \;\; 1 \leq i \leq s\} \subseteq \mathbf{K}^n$$

is called the **affine variety** of the ideal I.

The affine varieties are therefore the zero sets of polynomial equations. They describe the points which fulfill the kinematic equations. The objects of interest have been defined in a quite abstract way and now one needs methods of find their solution sets. The general situation in kinematics is as follows: the kinematic links together with the joints connecting them define a kinematic chain that imposes constraints to the end effector of the chain. The constraints can be expressed in polynomial form. A set of constraints forms a set of polynomials and generates, according to the definition above, an ideal. The solution set corresponding to the set of constraints is the affine variety belonging to this ideal.

As one has to be aware that the polynomials in the constraint equations are generally multivariate and solution algorithms depend heavily on multivariate divisions one needs at first some definitions on the term ordering in multivariate polynomials. The most important term ordering is given as an example.

Definition 4 (Lexicographic Order). Let $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $\beta = (\beta_1, \ldots, \beta_n)$ be elements of $\mathbb{Z}^n_{\geq 0}$. We define $\alpha >_{lex} \beta$ if the leftmost nonzero entry of the vector-difference $\alpha - \beta \in \mathbb{Z}^n$ is positive.

In Maple the keyword for this ordering is `plex`, e.g. a possible lexicographic termorder for polynomials containing the unknowns $\{x_1, x_2, x_3\}$ could be `plex(x[3], x[1], x[2])`. There exist other termorders like Graded Lex Order (Maple: `tdeg`) or Graded Reverse Lex Order. There is one more important termorder which is used in elimination processes:

Definition 5. lexdeg($L[1], ..., L[k]$**)** is an elimination order. The L[i] must be disjoint lists of variables. The variables in $L[i]$ are eliminated from the subsequent variables for all $i = 1..k - 1$.

This order is often used to eliminate variables when one does not want to compute an entire "plex" basis. It is equivalent to a product order that uses "tdeg" on each L[i]. The last ingredients necessary to define Gröbner bases is the notion of leading monomial, leading coefficient and leading term: A Gröbner basis is a very special generating set of the ideal I with some nice properties. Such a basis is not unique due to the fact that adding another polynomial to the generators does not change this property. Another reason for non-uniqueness is the fact that a monomial order has to be chosen first. One of the nicest properties of a Gröbner basis computed with a lexicographic term order ("plex" in Maple) is that it has the so called elimination property. To explain this property consider the following simple example:

Example 1. *The ideal* $I = \langle x_1^2 x_2 + x_1 x_2^2 + x_2^2 x_3, x_1 x_2 - x_3, x_2 x_3^2 - 1 \rangle$ *is given with the Gröbner basis wrt.* `plex(x[1],x[2],x[3])`

$$G = \langle x_3^7 + x_3^2 + 1, x_3^5 + x_2 + 1, -x_3^3 + x_1 \rangle$$

The first entry of G *is a univariate polynomial in* x_3 *and the remaining two polynomials contain the other two unknowns linearly, which means after solving the first polynomial for* X_3 *the back substitution becomes linear in the other variables. The univariate will always be in the unknown which is last in the sequence* $plex(x_i, x_j, x_k), i, j, k = 1, 2, 3$. *Note that the back substitution is not necessarily linear, depending on the geometric structure of the set of starting equations.*

4 Examples

4.1 Four-Bar Coupler Curve

In this example a planar four-bar is given with base points A_1, A_2, coupler points a_1, a_2 and the two arm lengths r_1, r_2 (Fig. 1). The question is to compute the coupler curve of a point $E(x, y)$ of the coupler system $(a_1; x, y)$. There are classical methods to do this, e.g. using complex numbers [9, pp. 66–69] or

Cartesian coordinates [8, pp. 178–180]. Here we use the kinematic mapping and the Gröbner basis to obtain the equation of the coupler curve. In the Maple worksheet below the coupler curve for the design values of Fig. 1 is computed $(A_1(0,0), a_1(0,0), A_2(5,0),\ a_2(3,0), r_1 = 3, r_2 = 4,\ E(2,1))$. T defines the general transformation matrix between the coupler system and the base. The two equations $gl1$ and $gl2$ say that the distances between points A_i and a_i are constant. $gl3$ and $gl4$ are equations coming from the transformation of point E to the base. Therefore one has four equations $gli, i = 1, ..4$. As the motion coordinates are homogeneous a homogenizing equation is added to a system of five polynomial equations. The Maple command `lexdeg([x0, y1, y2, x3],[x, y])` computes a Gröbner basis in which the first equation contains only the variables x, y and yields therefore the sought equation of the coupler curve. Note that the same procedure can be run without assigning values to the design variables, leading to the general equation of a coupler curve.

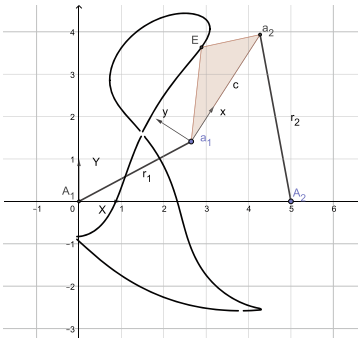


Fig. 1. Four-bar with coupler curve

Maple worksheet for the computation of the coupler curve

```
> T:= Matrix(3,3,{(1,1)=1,(1,2)=0,(1,3)=0,
> (2,1)=(-2*x0*y1+2*x3*y2)/(x0^2+x3^2),(2,2)=(x0^2-x3^2)/(x0^2+x3^2),
> (2,3)=-2*x0*x3/(x0^2+x3^2),(3,1)=(-2*x0*y2-2*x3*y1)/(x0^2+x3^2),
> (3,2)=2*x0*x3/(x0^2+x3^2),(3,3)=(x0^2-x3^2)/(x0^2+x3^2)}):
> A1:=Vector([1,0,0]):
> A2:=Vector([1,5,0]):
> a1:=Vector([1,0,0]):
> a2:=Vector([1,3,0]):
> gl1:=op(3,factor(numer(Transpose(A1-T.a1).(A1-T.a1)-3^2))):
> gl2:=op(3,factor(numer(Transpose(A2-T.a2).(A2-T.a2)-4^2))):
> b1:=Vector([1,1,2]):
> B:=Trans.b1:
> gl3:=numer(normal(B[2]-x)):
> gl4:=numer(normal(B[3]-y)):
> F:=[gl1,gl2,gl3,gl4,x0^2+x3^2-1]:
> bas:=Groebner[Basis](FF,lexdeg([x0,y1,y2,x3],[x,y])):
```

The first entry in the Gröbner basis *bas* is the implicit equation of the coupler curve:

$$\begin{aligned} &9\,x^6 + 27\,x^4y^2 + 27\,x^2y^4 + 9\,y^6 - 150\,x^5 - 30\,x^4y - 300\,x^3y^2 - 60\,x^2y^3 \\ &-150\,xy^4 - 30\,y^5 + 722\,x^4 + 300\,x^3y + 844\,x^2y^2 + 300\,xy^3 + 122\,y^4 - 570\,x^3 \\ &-330\,yx^2 - 570\,xy^2 - 330\,y^3 - 1151\,x^2 - 600\,yx + 49\,y^2 + 120\,x + 1320\,y + 800 = 0 \end{aligned} \tag{1.4}$$

Remark 1. *Fig. 1 was generated with Geogebra [1], a free dynamical 2d and 3d software package. The students can draw the four-bar linkage and animate it. The coupler curve equation can be generated in Geogebra using either the animation and the* `Locus` *command which might give an incomplete curve or the* `ImplicitCurve(f<xy>)` *command using the Eq. 1.2. This will produce the complete coupler curve. It is also possible to transfer the general coupler curve equation into Geogebra and play dynamically with the six design parameters* $(A_1, a_1, r_1, r_2, x, y)$ *of the four-bar linkage and the coupler point. With this the students can study the variety of different coupler curves. With the coupler curve equation also the double points and the focal circle can be found easily.*

4.2 Direct Kinematics of the Stewart Platform

In [5] the first algorithm to solve the direct kinematics (DK) of general Stewart-Gough platform (SGP) manipulators was presented. Using the kinematic constraint equations that comprise the given distances between the base and platform anchor points it was shown that after a sophisticated elimination process the final univariate polynomial in one of the unknowns could be computed. After numerical solving of this polynomial in a relatively complicated back substitution the remaining pos variables were found. This process was to complicated to be taught in a basic robotics course. The Gröbner basis algorithm which is now implemented in most of the symbolc manipulation software systems simplifies this task such that the DK of SGP can be taught even in a first robotics course. Due to lack of space a commented Maple worksheet is presented to show how the algorithm can be performed. Some of the display is truncated to fit in the paper. The design parameters used in this example are taken from the lower SGP of the hexapod telescope [6].

Maple worksheet: DK of the hexapod telescope manipulator.

T is the algebraic Transformation matrix between the platform and the base of the manipulator (see Fig. 2)

```
> T:=Matrix(4,4,{(1,1)=1,(1,2)=0,(1,3)=0,(1,4)= 0,
> (2,1) = (-2*x0*y1+2*x1*y0-2*x2*y3+2*x3*y2)/(x0^2+x1^2+x2^2+x3^2),
> (2,2) = (x0^2+x1^2-x2^2-x3^2)/(x0^2+x1^2+x2^2+x3^2),
> (2,3) = (-2*x0*x3+2*x1*x2)/(x0^2+x1^2+x2^2+x3^2), (2, 4) = (2*x0*x2+2*x1*x3)/(x0^2+x1^2+x2^2+x3^2),
> (3,1) = (-2*x0*y2+2*x1*y3+2*x2*y0-2*x3*y1)/(x0^2+x1^2+x2^2+x3^2),
> (3,2) = (2*x0*x3+2*x1*x2)/(x0^2+x1^2+x2^2+x3^2),
> (3,3) = (x0^2-x1^2+x2^2-x3^2)/(x0^2+x1^2+x2^2+x3^2), (3,4)= (-2*x0*x1+2*x2*x3)/(x0^2+x1^2+x2^2+x3^2),
> (4,1) = (-2*x0*y3-2*x1*y2+2*x2*y1+2*x3*y0)/(x0^2+x1^2+x2^2+x3^2),
> (4,2) = (-2*x0*x2+2*x1*x3)/(x0^2+x1^2+x2^2+x3^2), (4,3) = (2*x0*x1+2*x2*x3)/(x0^2+x1^2+x2^2+x3^2),
> (4,4) = (x0^2-x1^2-x2^2+x3^2)/(x0^2+x1^2+x2^2+x3^2)}):
```

A_1 are the coordinates of the base anchor points (centers of the spherical joints), a_1 are those of the platform anchor points in platform coordinate system and B_1 are the coordinates of b_i measured in the base system.

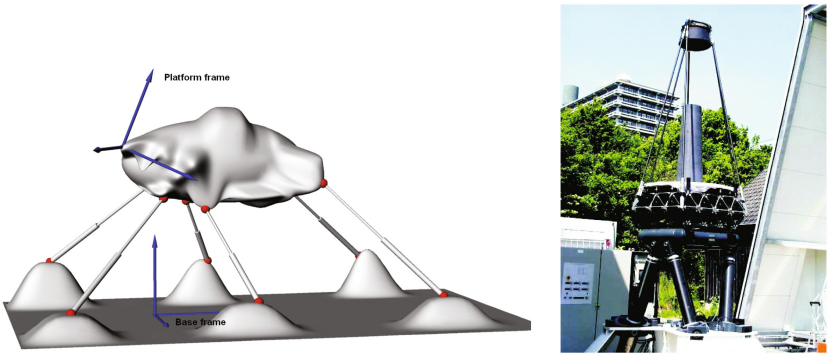


Fig. 2. (a) Stewart-Gough Platform Manipulator (b) Hexapod telescope

```
> for i from 1 to 6 do
> A||i:=Vector([1,a1||i,a2||i,a3||i]);
> b||i:=Vector([1,b1||i,b2||i,b3||i]);
> B||i:=T.b||i; od:
```

h_i are the distance constraint polynomials, r_i the given leg lengths and by adding 4 time the square of the Study quadric (Eq. 1.1) h_i become quadratic. This is allowed because of Definition 1.

```
> for i from 1 to 6 do
> h||i:=op(2,factor(numer(Transpose(A||i-B||i).
> (A||i-B||i)-r||i^2)+4*StudyQuad(stud)^2));
> od:
> a16:=a11:a15:=a12:a14:=a13:a26:=-a21:a25:=-a22:a24:=-a23:
> b16:=b11:b15:=b12:b14:=b13:b26:=-b21:b25:=-b22:b24:=-b23:
> a31:=0:a32:=0:a33:=0:a34:=0:a35:=0:a36:=0:b31:=0:b32:=0:
> b33:=0:b34:=0:b35:=0:b36:=0:
```

here the design parameters of the manipulator are set

```
> a11:=752:a12:=-139:a13:=-613:a21:=274:a22:=788:a23:=514:
> b11:=426:b12:=235:b13:=-661:b21:=517:b22:=627:b23:=110:
> r1:=730:r2:=727:r3:=719:r4:=721:r5:=737:r6:=741:
```

F is the definition of the ideal consisting of 8 polynomials in 8 unknowns, $x0 = 1$ is used as normalization of the homogeneous variable and again Definition 1 is applied, because differences of constraint equations are linear in the unknowns y_i (see [5])

```
> F:=[x0-1,h1,h2-h1,h3-h1,h4-h1,h5-h1,h6-h1,StudyQuad(stud)];
```

$$F := [x0 - 1, -367575\, x0^2 + 1088240\, x0\, x3 + 1304\, x0\, y1 - 972\, x0\, y2 + \ldots]$$

It makes sense to compute at first a total degree (tdeg) basis, because this usually is faster than a plex basis

```
> tdegBas:=Groebner[Basis](F,tdeg(y0,y1,y2,y3,x0,x1,x2,x3)):
```

From the tdeg basis a plex basis is computed in less than 1s on a notebook computer. The sequence of the unknown is important! The basis consists of eight polynomials.

```
> plexBas:=Groebner[Basis](tdegBas,plex(y0,y1,y2,y3,
x0,x3,x2,x1)):
```

Now the numerical part starts, one can set the accuracy to any digit wanted

```
> Digits:=40:
```

the first entry of the basis is the univariate which is solved for its unknown

```
> Lx1:=fsolve(op(1,plexBas),x1);
```

$$Lx1 := -0.49289752572694..., -.492573813561538..., \ldots$$

the first solution is inserted in the remaining 7 polynomials of the plex basis, which are then solve linearly! for the remaining variables

```
> LX1:=eval(plexBas,x1=Lx1[1]):
> fsolve({LX1[2],LX1[3],LX1[4],LX1[5],LX1[6],LX1[7],LX1[8]},
{x0,x2,x3,y0,y1,y2,y3});
```

$$\{x0 = 1.0, x2 = -0.271.., x3 = -0.00084.., y0 = 5.0159.., y1 = 109.1718.., y2 = -179.753..\}$$

5 Conclusions

The most universally useful aspect of applying Gröbner bases as a pedagogical tool is that with it the student has freedom to explore all problems that can be formulated as simultaneous polynomial equations. Thus one obtains clear insight into underlying structure and how a particular apparently novel unrelated problem is intimately but unexpectedly connected to a host of situations that have become familiar through previous study. This freedom is provided because the tool is so convenient and intuitively easy to use.

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